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**The use of blended teaching to improve workplace training in the transport  
sector**

**by**

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## DEDICATION

There is a time in one's life where reflection brings profound memories. My mind flash back to the greatest lessons I learnt from late grandmothers Musandiwa Mulaudzi and Tshinakaho Madzuvhana and my late father Reckson Nthangeni Sigwavhulimi Mulaudzi. I dedicate this dissertation to the above giants. My grandmother Musandiwa Mulaudzi had no benefit of education, however, she was a mathematician. She would divide a loaf of bread equally to her grandkids and no one will claim to have received a bigger or small portion. She could simplify, solve, and prove. Growing up, I thought I was her favourite and so were all her grandkids. She taught me to be humble. She never knew how to read, but at the end of the year she will demand to see a report. My grandmother Tshinakaho Madzuvhana also did not have the benefit of education and yet you could never rob her of change. She was a historian and a very great storyteller. She knew how to reference and would quote you verbatim. Each time I visit, she would ask if I had passed.

To my father, Reckson Nthangeni Mulaudzi, your love for education was unmatched. You passed matric when I was already in secondary school. Your matric certificate was so majestically hanged on the wall served as a constant reminder that you love education and take pride in your educational achievements. A book was always a part of your life, it did not matter what language they wrote the book in, it will meet your hands. You taught me the value of education. Your words were "education is an inheritance" (Pfunzo ndi ifa). No wonder my friends used to call you professor. You were a man of few words, but the power of your words was plentiful. You did not have a university degree, but you were a well of knowledge and wisdom to which our thirst was quenched.

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I am grateful to Passenger Rai Agency of South Africa for granting me permission to conduct the research.

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I would like to thank God for the blessing through the pandemic and for keeping me healthy and alive.

## ABSTRACT

The workplace environment is rapidly changing because of emerging technologies and access to the internet. It is indisputable that training is the propeller of productivity in the workplace. The fact that technical education in South Africa started in the railway sector is an indication that the entire transport sector values the importance of workplace training (Mathebula, Poya & Makwela, 2014). This study focuses on the use of blended learning teaching to improve workplace training in the transport sector. A questionnaire was mailed to Senior Technical Training Officers (STTO), drivers, train control officers, metro guards, yard officials and training managers in the transport sector to measure their views with respect to the use of blended learning in the transport sector in South African. The respondents see blended learning as satisfying and helps with time management. Reliable in connectivity and experience in using the learning could go a long way in ensuring the right blend of learning. The South African transport environment has always conducted workplace training in order to provide their employees with the necessary knowledge to do the job. To meet the demand of the modern world, the transport industry has also adopted Information and Communication Technology (ICT) supported training. According to Susilawati, (2015:8) the revolution in ICT has changed teaching and learning development and delivery. We are at the cusp of a new era where blended training must take priority over traditional training systems and this study seeks to also gather ideas for best practices when delivering and implementing blended learning. In this paper, the author will graphically demonstrate how leaning and development managers benefits from investing in a reliable connectivity and quality learner management system that responds to the needs of workplace learning.

**Key words:** Blended learning, information and communication technologies, online learning; hybrid instruction

## TERMINOLOGY USED IN THIS STUDY

The following terms are used in this study.

- The term Blended Learning, as used in this study, refers to “a modern educational technology, which is based on the concept of uniting the traditional face-to-face technologies and e-learning technologies based on new didactic opportunities of ICT and modern teaching means” (Shurygin & Sabirova, 2017).
- E-learning: learning that is facilitated by any electronic means. This is a very broad term that encompasses every use of electronic communication technology for example email, which leads to the facilitation of the learning process (Heinze, 2008).
- Mobile learning refers to the use of mobile or wireless devices for the purpose of learning while on the move.
- Train Control Officer - the employee on duty at a station or a Train Control Office who is in charge of and responsible for the working of trains and the operation of train control instruments and/or signals and points;
- Train Driver means the employee on duty in charge of and responsible for the working of a locomotive or motor-powered vehicle;
- Section Manager is the employee whose primary duty is to supervise and perform task observation on safety critical grade (SCG) employees as well as provide guidance to personnel in the operational tunnel.

## ABREVIATION AND ACRONYMS

|       |                                              |
|-------|----------------------------------------------|
| ETDP  | : Education Training & Development Practices |
| HR    | : Human Resources                            |
| LMS   | : Learner Management System                  |
| PRASA | : Passenger Rail Agency of South Africa      |
| QCTO  | : Quality Council for Trade and Occupation   |
| SCG   | : Safety Critical Grade                      |
| STTO  | : Senior Technical Training Officer          |
| TCO   | : Train Control Officer                      |
| TETA  | : Transport Education Training Authority     |



## Contents

|                                                                         |     |
|-------------------------------------------------------------------------|-----|
| DECLARATION .....                                                       | i   |
| DEDICATION .....                                                        | ii  |
| ACKNOWLEDGEMENTS .....                                                  | iii |
| ABSTRACT .....                                                          | iv  |
| TERMINOLOGY USED IN THIS STUDY .....                                    | v   |
| ABBREVIATION AND ACRONYMS .....                                         | vi  |
| LIST OF TABLES .....                                                    | ix  |
| CHAPTER 1. INTRODUCTION.....                                            | 1   |
| 1.1. THE NEED TO CONDUCT THE RESEARCH .....                             | 3   |
| 1.2. PROBLEM STATEMENT.....                                             | 4   |
| 1.3. AIM OF THE STUDY.....                                              | 5   |
| 1.4. THE RESEARCH DESIGN .....                                          | 5   |
| 1.5. PARTICIPANTS IN THE STUDY .....                                    | 6   |
| 1.6. THE RESEARCH STRATEGY.....                                         | 6   |
| 1.7. DATA COLLECTION .....                                              | 7   |
| 1.8. THE RESEARCH PROGRAM .....                                         | 7   |
| 1.9. SUMMARY .....                                                      | 8   |
| CHAPTER 2. LITERATURE REVIEW .....                                      | 9   |
| 2.1 INTRODUCTION.....                                                   | 9   |
| 2.2 The evolution of blended learning .....                             | 10  |
| 2.3 The origin of blended learning.....                                 | 10  |
| 2.4 Using the World Wide Web for blended learning .....                 | 11  |
| 2.5 The need for blended learning .....                                 | 13  |
| 2.6 The need for blended learning: The South African perspective .....  | 13  |
| 2.7 Greater access to knowledge and the need for blended learning ..... | 14  |
| 2.8 Advances in technology and the need for blended learning.....       | 14  |
| 2.9 Benefits of blended learning .....                                  | 15  |

|                   |                                                                                   |           |
|-------------------|-----------------------------------------------------------------------------------|-----------|
| 2.10              | Limitations of blended learning.....                                              | 16        |
| <b>CHAPTER 3.</b> | <b>RESEARCH METHOD .....</b>                                                      | <b>18</b> |
| 3.1.1             | THE PARTICIPANTS .....                                                            | 18        |
| 3.1.2             | Justification for conducting an exploratory study .....                           | 19        |
| 3.4.1             | QUESTIONNAIRES .....                                                              | 22        |
| <b>CHAPTER 4.</b> | <b>RESEARCH FINDINGS.....</b>                                                     | <b>26</b> |
| 4.1.1             | Analysis .....                                                                    | 26        |
| 4.1.2             | Findings.....                                                                     | 28        |
| 4.1.3             | OPEN ENDED QUESTIONS.....                                                         | 35        |
| <b>CHAPTER 5.</b> | <b>SUMMARY, DISCUSSION, RECOMMENDATIONS,<br/>CONCLUSIONS AND FINAL WORD .....</b> | <b>37</b> |
| 5.1               | LIMITATIONS OF THE STUDY .....                                                    | 38        |
| 5.2               | CONCLUSIONS EMANATING FROM THE FINDINGS .....                                     | 38        |
| 5.2.1             | Impact of blended learning on business efficiency .....                           | 39        |
| 5.2.2             | Flexibility and timely training.....                                              | 39        |
| 5.2.3             | Level of motivation towards blended learning.....                                 | 39        |
| 5.2.4             | Barriers towards the use of blended learning .....                                | 40        |
| 5.3               | RECOMMENDATIONS FOR FUTURE STUDIES .....                                          | 40        |
| 5.4               | conclusion .....                                                                  | 41        |
| 5.5               | A final word.....                                                                 | 42        |
|                   | References.....                                                                   | 43        |
|                   | Appendix A: Research request letter .....                                         | 47        |
|                   | Appendix B: Research Consent letters .....                                        | 48        |
|                   | Appendix C: Confidentiality agreement .....                                       | 49        |
|                   | Appendix D: REC application letter.....                                           | 50        |
|                   | Appendix E: Research questionnaire.....                                           | 51        |

## LIST OF TABLES

|                                                                                                  |    |
|--------------------------------------------------------------------------------------------------|----|
| Table 4.1: Definition of Likert scale points and related variables.....                          | 27 |
| Table 4.2: Definition of Likert scale points and related variables.....                          | 27 |
| Table 4.3: Definition of Likert scale points and related variables.....                          | 27 |
| Table 4.4: Definition of Likert scale points and related variables.....                          | 28 |
| Table 4.5: Length of time respondents have worked for their current employer .....               | 29 |
| Table 4.6: Length of time respondents have worked in construction.....                           | 29 |
| Table 4.7: Respondents' age .....                                                                | 30 |
| Table 4.8: Degree of concurrence relative to various statements .....                            | 31 |
| Table 4.9: Importance of the degree to which blended learning provides flexible training .....   | 33 |
| Table 4.10: Level of motivation towards blended learning.....                                    | 34 |
| Table 4.11: The degree to which barriers affect respondents towards using blended learning ..... | 35 |



## CHAPTER 1. INTRODUCTION

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***“No thief, however, skilful, can rob one of knowledge, and that is why knowledge is the best and safest treasure to acquire.” (Baum, 2017)***

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The workplace environment is rapidly changing because of emerging technologies and access to the internet. Lekchiri (2016) opined that technology, being one major element of the 21st century, changes clarify that the fast-technological advancements have shifted the workplace and changed the way we operate and learn. It is indisputable that training is the propeller of productivity in the workplace. The South African transport environment has always conducted workplace training in order to provide their employees with the necessary knowledge to do the job. To meet the demand of the modern world, the railway industry has also adopted Information and Communication Technology (ICT) supported training. According to Susilawati, (2015:8) “The revolution in Information and Communication Technology has changed teaching and learning development and delivery.” Although traditional face-to-face interaction is still the preferred mode of knowledge transfer, the transport sector recently began to use a Learner Management System (LMS) in an effort to incorporate both ICT training and the traditional face-to-face training. This has helped the organisations to look at training with a dual mind regarding learning and development.

We are at the cusp of a new era where blended training must take priority over traditional training systems. The fact that technical education in South Africa started in the railway sector is an indication that the entire transport sector values the importance of workplace training (Mathebula, Poya & Makwela, 2014). The adoption of ICT training in this environment to train technical skills must be encouraged. Presently, training programs are the cornerstone of initial, and refresher training for train drivers and train control officers (TCO) within PRASA. Train drivers and TCOs are recruited through human resources (HR) recruitment and selection processes. Successful candidates enter into “trainee status” for the duration of the training period which takes

up to 12 months. The training courses are accredited by the Quality Council for Trade and Occupation (QCTO) and Transport Education and Training Authority (TETA).

Theoretical training is conducted by qualified senior technical training officers (STTO) who have five or more years' experience in train operations. STTOs are registered as assessors and moderators with TETA. Practical on the job training is also conducted by a hand pick experienced accredited personnel. All employees who are operational, including managers who still engage with safety critical duties, are required to attend regular training.

There are many definitions for blended learning. Simply put, blended learning is the integration of traditional classroom learning experiences with online learning experiences. With the implementation of this teaching approach, there are many possibilities to apply learning and understanding in both environments (Kastner, 2019). While on-the-job training is a method that has been traditionally practiced, the requirement for facilitating the move of employees to a productive state quickly has decreased the amount of time and the quality of information the employee has to learn. (Callaway, 2005).

Gamble (2005) argues that, blended learning takes advantage of the power of technology to deliver training virtually. The integration of the internet and new technology brings about workplace training approaches that are flexible and accessible to employees. Boelens, Voet, & De Wever (2018) state that blended learning serves as an added advantage for greater flexibility in terms of time usage and accommodates a diverse student population beyond the physical environment of the institution.

Blended learning is fast becoming a key instrument in improving workplace training by employing a hybrid approach". Establishing a program that incorporates blended learning with a focus of improving the transfer of training is one-step towards improving student and workplace learning outcomes. "The opportunity of thoughtful combination of three delivery environments such as F2F synchronous in the traditional classroom (same time/same place), F2F synchronous as a live virtual classroom (same time/different place), and asynchronous (different time/different place), gives blended

learning advantages in comparison to pure traditional or distance learning environment (Prohorets & Plekhanova, 2014)

Traditional classroom and e-learning have their strength and weakness. The sensible thing to do will be to blend their strengths into blended learning. In this sense, it may be said that today the most natural form of learning is blended learning (Fong & Wang, 2007).

It is the aim of this study to explore how a blended learning approach can serve as an added advantage to workplace training. Responses provided by the participants in this study will be used to support or contradict the limited documented findings of recent studies in the field of blended learning in the workplace.

## **1.1. THE NEED TO CONDUCT THE RESEARCH**

The workplace is constantly changing the tools of the trade. The current virtual world requires employers to stay on track with all the changes that taking place. Mahesh and Woll (2018) explain that meeting the new demands requires an evolution from traditional learning to blended learning. Organisations should realize that the use of technology to teach can facilitate the integration of knowledge required by employees, who can help transfer the training to their workplace.

This study could provide an awareness for STTO and learning & development managers of exactly how blended learning can meet the various needs of workplace learners, specifically the ability to take charge of their learning. The findings from this study could motivate learning & development managers to invest in a quality LMS that responds to the needs of workplace learning, especially the use of simulators and videos. This study could also help the STTOs successfully implement blended learning in all their training. There is minimal research that explores how a blended learning approach can improve workplace training and this study could establish the degree to which blended learning provides flexible and timely training to enhance an employee-learning program in the transport sector. This study could suggest ways and means to adopt blended learning to improve workplace training.

Boelens, Voet and De Wever (2018) assert that blended learning provides opportunities for greater flexibility in terms of time usage and accommodation for a diverse student population outside of the physical walls of the institution. Workplace training should extend its scope to cater for the technologies that meet business and objectives. Truitt (2011) says the fact that there is a collaboration between training and the skills an employee must gain to better the probability of achieving the organisation's overall business and academic goals and objectives. Cheng and Chen (2015) state that the effectiveness of an e-learning system on, on-the-job training as opposed to traditional classroom training that applies to employers, especially focusing on the shift-duty staff.

## **1.2. PROBLEM STATEMENT**

The focal point of the study is the use of blended to improve workplace training in the transport sector. Research has shown that classroom-based training by itself is becoming too costly and difficult to manage. Organisations that embrace a blended learning model of training increase employee learning by utilising a hybrid of technology and face-to-face classroom facilitation (Snowden, 2009). Mahesh and Woll (2018) found that the employment of blended learning techniques at Intel has led to the reduction in tech time away from factory and the reduction of production tool downtime for training by 60% or more. They further concluded that overall, the impact data indicated that blended learning is a viable option and a cost-effective solution to provide support for equipment training at Intel from the view of the ROI and cost benefit analyses.

Kintu, Zhu and Kagambe (2017) argue that an effective blended learning environment is necessary in undertaking innovative pedagogical approaches through the use of technology in teaching and learning. Poon (2013) states that blended learning is particularly suitable for subjects that require hands-on experience, such as teacher training, construction technology, and property development.

Unfortunately, there is not much information about how instructors in the workplace use blended learning to provide safety critical training in the transport sector in South Africa environment and how the training can be personalized to correct or reset a fault.

### **1.3. AIM OF THE STUDY**

The aim of this study is to explore how a blended learning approach can be introduced in the transport sector with the view to improving worker competence.

In order to achieve this aim it is necessary to state the following objectives:

- To determine how the use of blended learning improves workplace training institution in the transport sector in South African.
- To establish the degree to which blended learning provides flexible and timely training to enhance an employee learning in the transport sector in South Africa
- To encourage the transport sector to adopt blended learning to improve workplace training
- To improve worker productivity by empowering workers to use blended training tools.
- To carve out a model for blending learning for the transport sector in South Africa

### **1.4. THE RESEARCH DESIGN**

An exploratory descriptive design involving a survey has been chosen. The survey collected details of the current attitude, knowledge and experience including barriers and motivation of senior technical training officers, managers, drivers and TCO to establish whether blended teaching improves workplace training. The questionnaire was developed to evaluate the degree to which blended learning provides flexible and timely training for employee-learning in the transport sector. These questionnaires were divided into three parts. The first part is designed to gather information on respondents' attitude, knowledge and experience of blended learning. The second part is designed to assess the respondents' motivation towards blended learning, and the third part will deal with barriers to using blended learning. For the purpose of this study random probability sampling is selected. The method of data collection produced quantitative data that can be analysed by statistical methods.

## **1.5. PARTICIPANTS IN THE STUDY**

The participants in this study include senior technical training officers who conduct training, train drivers who drives trains and attend refresher and other training interventions, train control officers who operates signals and attend refresher training, metro guards who handle safeguarding of commuters and often attend classes for training interventions, bus driver who drives buses and required to attend classes for customer services, service driver who drives train personnel for variety of reasons including taking them to remote depots, yard official shunt and marshal trains in the yard and manager who is in charge of all mentioned personnel.

## **1.6. THE RESEARCH STRATEGY**

The strategy of this study can be specified as exploratory where quantitative data will be gathered. Bhat (2018) defined exploratory research as a research used to investigate a problem which is not clearly defined. In the case of my study, there is no prior scientific knowledge or research that suggests that the use of blended learning improves workplace training. There is not a single empirical study about blended learning in the South African transport sector. Theoretical studies about the use of blended learning to improve workplace training have also been limited and mostly confined to defining blended learning and describing blended learning approaches. The common comprehensive description of blended learning is that it is a mixture of technology and face-to-face classes.

The absence of a transport sector university or college that is open for the public in South African has brought limitations in railway/road workplace learning research. Senior technical training officers are qualified in Education Training and Development Practices (ETDP) in addition to their operational experiences. The purpose of this research is to explore how blended learning can improve workplace learning. As much as there are grey areas in workplace learning and as a result of the current pandemic, I am able to adapt to change as the research progresses without incurring any costs. Taking into consideration the new normal of social distancing, this research lays the foundation to further research.

The strategy of this study is quantitative as it involves the collection of data in a numeric form. The data will be collected by administering a questionnaire using probability sampling to give every member of my population a chance to be selected. What makes the population to be suitable is because STTOs are the ones responsible for conducting the training, Managers oversee the training and drivers and train control officers attend training. The questionnaire will be distributed electronically to the target population with instructions and their consent. For the processing and analysing of questionnaire data the Statistical Package for the Social Sciences (SPSS) will be utilised.

Inductive reasoning will be an integral part of the construction of both statements and relationships, as well as in the formulation of a conclusion. Inductive reasoning used in the interpretation of learner experiences will be used in setting up the list of issues that are relevant to the South African transport industry. Based on the data from the experiences of the participants, I will be able to conclude if the use of blended learning improves workplace training.

## **1.7. DATA COLLECTION**

A survey to get participants' feedback on the use of a blended learning will be conducted. Closed-ended descriptive questions where respondents choose among fixed responses to establish whether blended teaching improves workplace training has been compiled. STTO, drivers (train, truck & bus) will be approached randomly to take part in this study.

## **1.8. THE RESEARCH PROGRAM**

The subsequent chapters of this study will include the following:

- Chapter two contains the literature review that describes blended learning. It describes blended learning approaches in the workplace within a broader framework of knowledge and validates the research construct.

- Chapter three describes the research methodology employed in this study, starting with the context of the study and the research plan. Detail on sampling procedures and data collection will be provided.
- In chapter four, data analysis will be followed by the configuration and construction of the final data categories.
- Chapter five deals with the findings of the study. The usefulness of these findings will be contemplated. Conclusions will be drawn from the results to provide a logical ending to the study.

## **1.9. SUMMARY**

This chapter served as an overview of the research study. The background and rationale of the study was placed within a workplace learning environment. This was followed by formulation of the research problem and aims of the research. The design of the study as well as data collection and analysis were described briefly. It was established that to meet the demand of the modern world, the transport sector has also adopted ICT supported training. The experiences of the participants' blended environment will provide an information of whether a blended approach can improve workplace learning in the transport sector. The findings of the study may be unique to the transport sector but might be of interest to any other workplace environment.

## CHAPTER 2. LITERATURE REVIEW

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*“I mean, you could claim that anything's real if the only basis for believing in it is that nobody's proved it doesn't exist!” (Rowlings, 2019)*

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### 2.1 INTRODUCTION

Chapter 2 is a review of the literature supporting this research: The use of blended teaching to improve workplace training in the transport sector. The literature concentrates on the following areas:

- The revolution of blended learning;
- The origin of blended learning;
- Using the world wide web for learning;
- The need for blended learning;
- The need for blended learning: South African perspective;
- Greater access to knowledge and the need for blended learning;
- Advances in technology and the need for blended learning;
- Benefits of blended learning;
- Limitations of blended learning; and
- Possible Solutions for Blended learning Implementation.

## **2.2 THE EVOLUTION OF BLENDED LEARNING**

Zhang and Zhu (2017) observe that the term blended learning is being used with increased frequency in both academic and corporate circles, the authors envisage that blended learning could become one of the most significant developments of the 21st Century. In their research the present and future state of blended learning in workplace learning settings in the United States of America (USA), Kim, Bonk & Oh (2008) argue that blended learning has become a popular delivery mode in workplace learning settings. Blended learning is transforming the landscape of classrooms over the past few years, as technology has become more readily accessible. Jones (2019). According to Hilliard (2015) blended learning is a fast-growing trend in traditional institutions in higher education and other organisations.

Medina (2018) views blended learning as a pedagogical approach that offers educational communities the opportunity to customise their learning by using synchronous and asynchronous delivery modes to increase levels of interaction among the agents involved (generally, students and faculty members). Listiana and Jaharadak (2019) assert that practices of blended learning have become a norm in educational networks around the world, enabling students to enjoy all the more fascinating and effective instruction, which is tailored to their own needs.

## **2.3 THE ORIGIN OF BLENDED LEARNING**

There is an academic discourse as per the origin of blended learning. According to Güzer and Caner (2014) blended learning emerged as one of the most popular pedagogical concepts at the beginning of the year 2000. Friesen (2012). points out that the precise origin of the term blended learning is uncertain but acknowledges however, that one of the first occurrences that has been identified is its use in a 1999 news release from the Effective Practice Incentive Community (EPIC) Learning, an Atlanta based computer skill certification and software training business.

While (Bersin, 2003; Gamble, 2005; Marsh, 2001; Smith, 2001) concede that blended learning is a fairly new term in the industry, the concept has been around for many years and is one of the leading trends in the training industry today. Modern computer-based training can be traced back to the mini-computer and mainframe training of the

60's and 70's. It was the first time that training could be deployed to countless workers within an organisation without having to rely on printed materials and face-to-face instruction (Pappas, 2015).

Bersin (2004) draws our attention to the fact that the first technology training approach came with a mainframe and minicomputer in the 1960s and 1970s. These systems had the limitations of character-based terminals but the benefit of reaching hundreds to thousands of people at their workplace. A pioneering example of such system was Programmed Logic for Automatic Teaching Operations (PLATO), a system developed in 1963 by control data University of Illinois in the USA. PLATO pioneered the use of computers in a traditional educational setting and it still exists today. However, according to Güzer and Caner (2014) blended learning become the focal point at the beginning of the 2000 as an eclectic approach while scholars are debating if it is conducive for students. Research has shown that that blended learning is traced to be used first time at the beginnings of 2000 (Google Scholar, 2021). This led them to a conclusion that idea of blended learning has a scratchy history within the recent twelve years. However, Holt (2011) argues that blended is not something new it is an art that has been practised by inspirational teachers such as Judi Krishnamurti for centuries. It is notable that over the past few decades, blended learning has amplified learning methodologies, opening up new opportunities for open and distance learning as well as challenging old paradigms.

## **2.4 USING THE WORLD WIDE WEB FOR BLENDED LEARNING**

“Blended learning has been in use since the popular advent of the Internet and the World Wide Web in the late 1990s.” (Nuruzzaman, 2016). Singh (2021) explains that blended learning programs may include several forms of learning tools, such as real-time virtual/ collaboration software, self-paced Web-based courses, electronic performance support systems (EPSS) embedded within the job-task environment, and knowledge management systems.

Thomas (2010) acknowledges that in the world of pervasive Internet, and the Web 2.0 social networking technologies 2, learners are also evolving into a new genre—the so-called “digital natives” who want to be in constant communication with their peers, expect individualized instruction and a personalized learning environment, which

automatically adapt to their individual needs. This is a new tribe since a tribe is no longer defined by a geographical location but by the shared global mindset. The affordances of Web 2.0 seem to fit with modern thinking about the nature of learning for the information age and the corresponding educational practice expected by the 'digital native' students.

Educators realise that the www has become a source knowledge using internet Brooks, Nolan and Gallagher (2006:4) enhance this point that "The web is changing both what we teach and how we teach". The web provides a new learning environment and a wealth of pedagogic possibilities; thus, teachers and administrators are showing an increased interest in becoming part of this "new wave" of educators. (Yang 2001:156).

The advent of website influenced schools and universities to invest in Information Communication Technology (ICT) education. Ross and Schulz (1999) opine that "The World Wide Web, a sub-facet of the Internet, is often the sole reason schools and universities allocate millions of dollars for new computer networks and hardware". They also observe that "certainly, the Web as an educational technology has tremendous potential to support student learning. The same technology is motivating many college professors to re-evaluate the way they teach and altering the way students learn."

The Covid-19 pandemic has been the antecedence for the digitalisation of education and the internet has become the nucleus for teachers to collaborate. Singh, Zhu, Thakkar and Ravaioli, (1996) argue that "The WWW can provide an extremely useful platform for interactive learning and distance education". They also noted that the World Wide Web (WWW) on the Internet has been recognised as an effective environment to create new distributed applications that have the potential to bring instruction beyond the bounds of the classroom.

Ahmed, Shuja and Chaudhry (2014) state that rapid growths in digital technology and World Wide Web capabilities altered the playing field for almost all the stakeholders in the private and public sectors. They further established that this evolution enables entities to communicate with their audience. For example, social media and the World Wide Web can profit educational institutions, especially in developing countries to

effectively communicate with current and prospective students, faculty, parents, and donors.

## **2.5 THE NEED FOR BLENDED LEARNING**

Considering social distancing brought about by the Covid-19 pandemic, the need for blended learning has increased as it allows for learning to continue virtually. Bonk & Graham (2012) argue that blended learning serves as an added benefit due to the emergence of geographically dispersed work environment in which personnel is constantly being changed. Tayebinik and Puteh (2013) opine that the blended learning approach is likely to emerge as the predominant instructional model in the unforeseeable future. They further argue that those activities which are involved in the blended instruction can foster the sense of community belonging and remove the frustration created by mere online environment.

McKenzie, Perini, Rohlf, Toukhsati, Conduit and Sanson, (2013) assert that blended learning approaches as shown by research have amplified the need for school leaders, teachers, students and the wider community to take advantage of learning opportunities afforded through improved personalisation, collaboration, and communication enabled by learning technologies.

## **2.6 THE NEED FOR BLENDED LEARNING: THE SOUTH AFRICAN PERSPECTIVE**

According to Basitere and Ivala (2017) the emergence of the ‘# Fees Must Fall’ protests in the South African higher education institutions has necessitated discussions on the improvement of the implementation of blended learning to complement the traditional campus face-to-face lecture delivery. The authors further point out that while blended learning might be seen as a success in some developed countries, its implementation in developing or third world countries such as South Africa, where there is a differential access to resources enough caution should be exercised.

Olivier (2020) states that that blended learning is the only the only solution, to a country’s unequal teaching environment. It integrates computer-assisted online activities with the conventional face-to-face teaching (chalk-and-talk). It is notable that

in the South African context, mainstream blended learning is not the antidote to the educational challenges because there is a need to go beyond blended learning. The use of blended learning has increased in higher education internationally, and South Africa is no exception. However, the digital divide has widened as a result of various demographic factors, such as urban and rural geographical locations, affluent universities as well as historically disadvantaged universities (Zimba, Khosa & Pillay, 2021).

## **2.7 GREATER ACCESS TO KNOWLEDGE AND THE NEED FOR BLENDED LEARNING**

To expand teaching and learning in the workplace, training and development managers must make available additional tools that will clarify the intricacies of blended learning. The availability of tools will provide learners with access to quality education by allowing students to learn virtually. Dziuban, Graham, Moskal, Norberg and Sicilia (2018) argues that one of the added benefits of online technologies is the increase access to non-traditional and underserved students by bringing a host of educational resources and experiences to their doorstep.

Blended learning can also be deemed part of the continuing trends of non-traditional and distance learning. This school of thought zero in on making learning available to learners in a variety of delivery formats and adaptable to myriad styles or preferences. Kim *et al* (2006). Lungu (2013) who opines that blended-learning courses provide the learners with a real chance to experience independent learning and it is a truism that a blended-learning model is a right step towards lifelong learning.

## **2.8 ADVANCES IN TECHNOLOGY AND THE NEED FOR BLENDED LEARNING**

Singh (2021) is of the view that while learning technologies and delivery media continue to evolve and progress, one thing is certain: organizations (corporate, government, and academic) are in favour of blended learning models over single delivery mode programs. Vickers (2017) asserts that the ability to know the potential technology is half the struggle. On the other hand, deciding how best to apply these technologies in a classroom setting is another half struggle. The advancements in

technology have created exciting new possibilities in the world of education (Crawford & Jenkins, 2017).

In this digital age, blended learning has grown into a popular trend that uses the digital platform for educational purposes (Rosenbusch 2020). Graham (2009) points out that technological advances and accessibility to ICTs facilitate the development of blended learning approaches in both higher education and corporate training contexts. However, technological advances through innovative software programs is open for effective and powerful synchronous collaborations among a community of learners who may be dispersed across the country and beyond (Groen & Li 2005).

## **2.9 BENEFITS OF BLENDED LEARNING**

Grassian, Botello, Phares and Turnbow (2005) state in the University of California Los Angeles (UCLA) Blended Instruction Course (BICo) Task Force Report that blended instruction is a viable choice, given limited staff as well as very large and growing learner populations”. They came up with some of the practical benefits of blended instruction:

- The ability to share, standardize and reuse content;
- Learner control over pace and sequencing, and the ability to review online content multiple times;
- Immediate feedback on computer-graded quizzes and exercises and quicker feedback on writing assignments;
- Reduced grading load for instructors;
- Lessened demand for classroom space; and
- More flexible scheduling for students and instructors.

Poon (2013) pronounce that blended learning influences students’ perceptions of their learning ecosystem and their academic achievement”. The author advances the point by explaining that blended learning benefits students and institutions. The author further adds that blended learning facilitates improved learning outcomes, access

flexibility, a sense of community, the effective use of resources, and student satisfaction.

In a white paper entitled “Achieving Success with Blended Learning”, Singh and Reed (2001) explain; the concept of blended learning is rooted in the idea that learning is not just a one-time event, but that learning is a continuous process. They added that blending provides various benefits over using any single learning delivery type alone:

- Improved learning effectiveness;
- Extending the reach;
- Optimizing development cost and time.

## **2.10 LIMITATIONS OF BLENDED LEARNING**

Al Zumor, Al Refaai, Eddin and Al-Rahman (2013) argue that the salient limitations for blended teaching were Internet connectivity and the technical problems encountered by Blackboard users. The research shows that there is a natural tendency and preference of learners to use books rather than technology, the problem of cheating facilitation, the difficulty of Blackboard's inbuilt or teachers' instructions, social isolation, computer illiteracy, and access to private computers. All of these limitations occur in the uncertainty category.

According to Kim *et al* (2006) fast changing technology and insufficient management support and commitment are viewed as the most significant issues that must be addressed to implement blended learning successfully. Also, thirteen percent of our respondents viewed the lack of understanding of what blended learning was as the most significant barrier to implementing it. According to Rossett and Frazee (2006), blended learning provides hope to those who want to move beyond the limitations of classroom instruction and training schedules – too often experiences that are set in time and bounded by place”.

Easy access to online information and advances in technology make it possible to resolve these limitations by adopting blended learning approaches. Blended learning strategies have been shown to improve students' academic performance, motivation, attitude, and satisfaction, and to provide convenient and flexible learning.

Implementation of blended learning strategies has also proved cost effective (Khalil, Abdel Meguid & Elkhider 2018)



## **CHAPTER 3. RESEARCH METHOD**

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*“Sometimes it is the methods of learning, more so, than what is learned, that is useful!”*  
(McCoy, 2019).

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### **3.1 OVERVIEW**

After the literature review, the researcher discusses the research methods used during the research process. Abu-Taieh, Al Hadid, and El Mouatasim (2020) establish that research methodology is the path through which researchers need to conduct their research. The researcher followed this path to gather the information and to determine what to do with the information collected. This chapter also describes the research design that the researcher followed to collect, record and analyse data in this study.

#### **3.1.1 THE PARTICIPANTS**

The participants in this study include senior technical training officers who conduct training, train drivers who drives trains and are required to attend refresher and other training interventions, train control officers who operates signals and attend refresher training, metro guards who handle safeguarding of commuters and often attend classes for training interventions, bus driver who drives buses and required to attend classes for customer services, service driver who drives train personnel for variety of reasons including taking them to remote depots, yard official shunt and marshal trains in the yard and manager who is in charge of all mentioned personnel.

### **3.2 THE RESEARCH DESIGN**

Heppner, Wampold, Owen and Wang (2015) describe research design as the tool that researchers use to frame a study and involves developing a plan or procedure to investigate. Design is defined by Akhtar (2016) as drawing an outline or arranging or planning details. It is the method of building decisions before a situation arises in which the decision has to be carried out.

This research study is characterized as an exploratory-descriptive study that uses a quantitative approach. The researcher justifies why an exploratory descriptive quantitative approach was the most suitable for this study below.

### **3.1.2 Justification for conducting an exploratory study**

The researcher observes that workplace training in the transport sector is taking place and conducted by in-house trainers and seeks to understand more about blended learning to teach. This illuminates the researcher's desire to travel through this uncharted terrain of blended learning in the railway training environment to explore if this observation can be explained by existing theories. Stebbins (2001) explains that exploratory research is the action of putting one's self deliberately in place again and again where discovery is possible and broad. Simply put, exploratory research can be described as the act of laying the initial groundwork for future research. The researcher takes a step further and attempts to explain the use of blended teaching to improve workplace training in the transport sector.

The researcher is now driven to describe what is happening within blended learning in the railway environment in more detail, adding the missing parts and expanding the understanding by collecting required information. This means the researcher can create a fuller picture of what blended learning is. Akhtar (2016) opines that descriptive research aims at portraying accurately the characteristics of a particular group or situation and further posits that it is used to identify and get information on characteristics of a particular issue like community, group or people.

The researcher aims to investigate how the use of blended teaching can improve workplace training in the transport sector. Elman, Gerring, & Mahoney (2020) define exploratory research as a study comprising an attempt to discover something new and interesting, by making your way through a research topic. They also point out that exploratory research is a soul of excellent research.

According to Zikmund, Ward, Lowe, Winzar and Babin (2011) researchers explore when they have little or no scientific knowledge about the group, process, activity, or situation they want to examine but have reason to believe it contains elements worth discovering. The researcher seeks to clarify the nature of blended learning in the

railway institution and to lay a good foundation for future studies regarding training in the railway workplace. Fisher, Laing, Stoeckel and Townsend (1991) draws our attention to the fact that exploratory study is needed whenever there is a perceived program problem, but the nature of the problem simply is not known. Zikmund, Ward, Lowe, Winzar and Babin (2011) posit that some exploratory research may be necessary to gain insights into such a problem.

Elman, Gerring and Mahoney (2020) argues that an exploratory study can essentially be carried out for two different purposes, to increase the knowledge of a topic that is little known but needs to be better known and to generate new and interesting hypothesis about a topic that is already known. Fisher, Laing, Stoeckel and Townsend (1991) maintains that exploratory studies rarely require hypothesis statements because they seldom test relationships between variables.

The researcher adopted exploratory in this study in order to gain insight into blended learning from the railway training perspective. This research will hopefully illuminate the need for further research on this topic or to adopt blended learning in their workplace training environment.

Bless, Higson-Smith and Kagee (2006) exploratory research is used to gain insight into a situation, phenomenon, community, or person.

Swedberg (2018) offers the goals and methods of different types of exploratory studies and aligned the study the two below:

- The standard exploratory study (type 1)

The goal is to explore a topic that is little known, and to produce a publishable work.

- The informal exploratory study (prestudy)

The goal is to develop new ideas; and any means whatsoever can be used. The size of the sample can be small or fair.

The need for this type of study arises from the absence of basic information on how transport sector learners, managers and trainers currently experience blended learning.

Akhtar (2016) observed descriptive research as a study that answers the questions, what, who, where, how and when. It is used to study the current situation and provide the researcher with detailed information about a phenomenon. Akhtar (2016) offer the following steps of Descriptive Research

- The Clear discrimination of objective of Research
- Determination of the Method of Data Collection
- Selection of Sampling
- Real Collection of Data
- Analysis of Achieved conclusion

### **3.3 RESEARCH METHODS**

The research method used in this study is a survey. Salaria (2012) explains that the survey research employs applications of scientific method by critically analysing and examining the source materials, by analysing and interpreting data, and by arriving at generalization and prediction. Gable (1994) observes that the survey approach refers to a group of methods which emphasize quantitative analysis, where data for many organizations are collected through methods such as mail questionnaires, telephone interviews, or from published statistics, and these data are analysed using statistical techniques.

A questionnaire comprising open-ended, closed-ended, Likert scales; was administered to respondents by mail. In order to determine whether questions are clear and understandable and also fit the purpose of extracting the type of information that they collected, the researcher first sought inputs of other individuals. According to Gable (1994) for a survey to explain causal relationships or even in providing descriptive statistics, it must contain all the right questions asked in the right way. The

researcher use the survey studies to information about respondent's behaviors, attitudes, and opinions.

### **3.4 METHODS OF DATA COLLECTION USED IN THIS STUDY**

This section summarizes research methods and data collection. Fischer (2013) defines research data as the recorded factual material commonly accepted in the scientific community as necessary to validate research findings. A discussion of the methods used will now follow.

#### **3.4.1 QUESTIONNAIRES**

Participants were requested to complete a questionnaire that focuses on their experience with blended learning (See appendix B). The aim of the questionnaire was to evaluate the degree to which blended learning provides flexible and timely training for employee-learning in the transport sector. This questionnaire is divided into three parts. The first part gathers information on respondents' attitude, knowledge and experience of blended learning. The second part assessed the respondents' motivation towards blended learning, and the third part will deal with barriers to using blended learning.

### **3.5 THE TRUSTWORTHINESS OF THE RESEARCH**

Neisse and Wegdam (2008) define trustworthiness as a measurement of the reliability of a context information provider to provide context information about a specific entity according to a certain quality level. The method used to establish trustworthiness in this study include internal validity, external validity, reliability, and objectivity. Jordan (2005) state that most fundamentally, research that is not trusted is worthless. Neisse, and Wegdam (2008) also believe that in its more general definition, trustworthiness is represented as a subjective measurement of belief from one entity regarding the behaviour of another entity focused on a certain trust aspect. The trustworthiness of this study is based on the strategies described in the paragraphs that follow.

### **3.6 CREDIBILITY OF THE FINDINGS**

In order for the researcher to ensure credibility of the research, questionnaires were shared with managers to distribute through emails and online platforms. Participants without computers could use managers' computers to complete the questionnaires. Korstjens and Moser (2018) points that credibility is the equivalent of internal validity in quantitative research and is concerned with the aspect of truth-value. The researcher aware that credible research is achieved by assessing whether the research findings represent a credible conceptual interpretation of the data drawn from the participants' original data. Kyngäs, Kääriäinen and Elo (2020) define credibility as confidence in the 'truth' of the findings.

#### **Data Collection**

Data for this research was collected by means of:

- A review of existing literature
- A survey questionnaire

Literature review play an important role in the research study and it goes beyond simply collecting data. Existing literature add value to the correctness of the research because it enables the researcher to gain understanding of area of study. The literature review guided the researcher to refine the topic and navigated a direction of the study. The survey question followed the path from the theory studied in the literature review. The pertinent questions are accessible in Appendix F.

#### **Data Recording**

The general purpose of data recording is to set in writing and assure the preservation of the data collected in field or laboratory studies (Sahn, Lockwood & Scrimshaw, 1984). The data recording method for this study was registering participants' responses to the spreadsheet and filling all questionnaires into a folder.

## **Data Analysis**

Wray, Watson, Gibson and Barrett (2020) state that quantitative research may be considered as a way of thinking about the world. It is essentially deductive: measurements are made, analysis is applied, and conclusions are drawn.

### **3.7 ETHICAL CONSIDERATIONS**

Ethics is defined by Etics, Ferreira and Serpa (2018) as a rigorous procedure of reflection on human action, with the purpose of questioning about its meanings, the reasons and values that motivate choices and their foreseeable consequences, to guide decisions that respect human dignity. The researcher informed participants that no harm or discomfort would be associated with their participation in the study.

Appendix A is a request letter used to apply for permission from the managers to conduct research in the railway operations. Upon receiving approval (see appendix B: Research Approval Letter), contact was made with the participants to forward questionnaires. The procedure followed for survey was explained to all participants and all questions regarding the research were answered. The researcher explained to the participants that they have rights to withdraw from the study at any stage if they wish to do so and the participants were asked to sign the Informed Consent forms.

All effort was made to ensure the participants felt comfortable, calm, and relaxed during the interviews. The participants were not discriminated against or stigmatised regarding their perceptions of the study. The confidentiality and privacy of all participants was protected throughout the research. All data captured through written notes and voice recordings were held in the strictest of confidence, ensuring real names and identity were not used in the research.

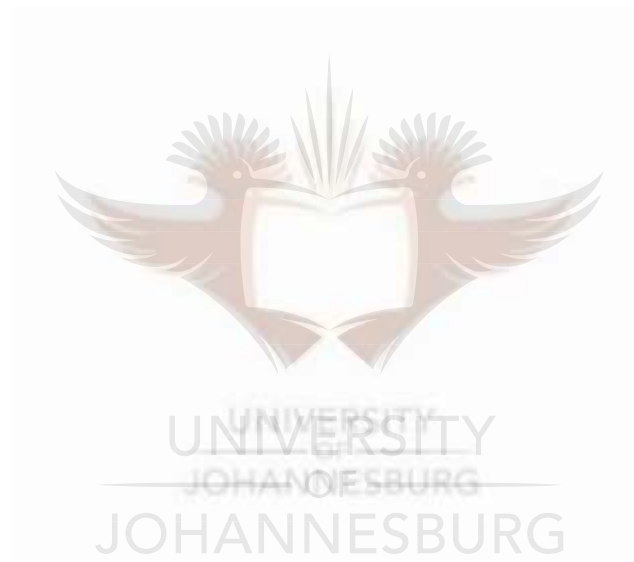
The researcher followed Gajjar (2013) ethical consideration in research;

- Honesty
- Objectivity
- Integrity
- Carefulness

- Openness
- Respects for Intellectual Property.

### **3.8 SUMMARY**

In conclusion, this chapter set out to clarify the selection and relevance of the research design and the methodology. This research adopted an exploratory descriptive approach. The research methodology detailed the data collection and data recording. Ethical considerations and a brief description of the participants were also included. The next chapter will present the findings from the data collection for the research.



## CHAPTER 4. RESEARCH FINDINGS

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*“The greatest part of a writer's time is spent in reading, in order to write: a man will turn over half a library to make one book”* (Johnson, 2008).

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Twenty three (23) Responses were received from a convenience sample of six (6) Senior Technical Training Officers, five (5) Train Drivers, three (3) Metro Guards, three (3) Train Control Officers, two (2) Bus Drivers, two (2) Service Drivers, one (1) Manager and one (1) Yard Official. A total of twenty-three (23) responses were included in the analysis of the data.

### 4.1.1 Analysis

The analysis of the data consisted of the calculation of descriptive statistics to depict the frequency distribution and central tendency of responses to fixed response questions to determine the degree of importance.

Close ended questions with five-point Likert scales, which also included an ‘unsure’ response option were used. Therefore, to rank fixed response items according to the central tendency of responses, mean scores (MSs) were calculated as follows:

$$MS = \frac{1n_1 + 2n_2 + 3n_3 + 4n_4 + 5n_5}{(n_0 + n_1 + n_2 + n_3 + n_4 + n_5)}$$

The variables are referenced in Table 1, 2, 3 and 4.

**Table 4.1: Definition of Likert scale points and related variables**

| <b>Likert scale point</b> | <b>Variable</b> |
|---------------------------|-----------------|
| Strongly Disagree         | n <sub>1</sub>  |
| Disagree                  | n <sub>2</sub>  |
| Neutral                   | n <sub>3</sub>  |
| Agree                     | n <sub>4</sub>  |
| Strongly Disagree         | n <sub>5</sub>  |

**Table 4.2: Definition of Likert scale points and related variables**

| <b>Likert scale point</b> | <b>Variable</b> |
|---------------------------|-----------------|
| Not at all important      | n <sub>1</sub>  |
| Slightly important        | n <sub>2</sub>  |
| Moderate important        | n <sub>3</sub>  |
| Very important            | n <sub>4</sub>  |
| Extremely important       | n <sub>5</sub>  |

**Table 4.3: Definition of Likert scale points and related variables**

| <b>Likert scale point</b> | <b>Variable</b> |
|---------------------------|-----------------|
| Not all motivating        | n <sub>1</sub>  |
| Slightly motivating       | n <sub>2</sub>  |
| Undecided                 | n <sub>3</sub>  |
| Very motivating           | n <sub>4</sub>  |
| Extremely motivating      | n <sub>5</sub>  |

**Table 4.4: Definition of Likert scale points and related variables**

| Likert scale point | Variable       |
|--------------------|----------------|
| Not affected       | n <sub>1</sub> |
| Slightly affected  | n <sub>2</sub> |
| Neutral            | n <sub>3</sub> |
| Very affected      | n <sub>4</sub> |
| Extremely affected | n <sub>5</sub> |

#### **4.1.2 Findings**

Table 5 indicates that 45.3% of respondents have worked for their current employer ' $> 1 \leq 5$ ' years, followed by 19% ' $> 5 \leq 10$ ' years, and 11.9% relative to each of ' $\leq 1$ ', ' $> 10 \leq 20$ ', and ' $> 20$ ' years. In summary, 88.1% of respondents have worked for their current employer  $> 1$  year, 42.8% for  $> 5$  years, and 23.8%  $> 10$  years. The mean length of time respondents have worked for their current employer is 7.9 years. Therefore, the respondents can be deemed to have had exposure to their organisations, which contributes to the reliability of the findings.

**Table 4.5: Length of time respondents have worked for their current employer**

| <b>Period<br/>(Years)</b> | <b>Response<br/>(%)</b> |
|---------------------------|-------------------------|
| $\leq 1$                  | 11.9                    |
| $> 1 \leq 5$              | 45.3                    |
| $> 5 \leq 10$             | 19.0                    |
| $> 10 \leq 20$            | 11.9                    |
| $> 20$                    | 11.9                    |

1% of respondents have worked in construction  $\leq 1$  year, followed by 14.6% ' $> 1 \leq 10$ ' years, 26.8% ' $> 11 \leq 20$ ' years, 34.2% ' $> 21 \leq 30$ ' years, and 22.0% ' $> 31 \leq 44$ ' years (Table 3). In summary, 97.6% of respondents have worked in railways  $> 1$  year, 83% for  $> 5$  years, and 56.2%  $> 10$  years. The mean length of time respondents have worked in construction is 14.5 years. This implies that the respondents can be deemed experienced, which contributes to the reliability of the findings.

**Table 4.6: Length of time respondents have worked in construction**

| <b>Period<br/>(Years)</b> | <b>Response<br/>(%)</b> |
|---------------------------|-------------------------|
| $\leq 1$                  | 2.4                     |
| $> 1 \leq 10$             | 14.6                    |
| $> 11 \leq 20$            | 26.8                    |
| $> 21 \leq 30$            | 34.2                    |
| $> 31 \leq 44$            | 22.0                    |

Table 7 indicates that 42.9% of respondents were ' $> 21 \leq 40$ ' years of age, followed by 23.9% ' $\leq 21$ ', 14.3% for each of ' $> 41 \leq 50$ ' and ' $> 51 \leq 63$ ', and only 4.8% ' $> 63$ '. In summary, 66.8% were ' $\leq 41$ ' years of age, and 33.2% were  $> 41$  years. Per definition relative to workers, people  $> 41$  years of age are 'older workers'. The mean age was 37.9 years.

**Table 4.7: Respondents' age**

| Period<br>(Years) | Response<br>(%) |
|-------------------|-----------------|
| $\leq 21$         | 23.9            |
| $> 21 \leq 40$    | 42.9            |
| $> 41 \leq 50$    | 14.3            |
| $> 51 \leq 63$    | 14.3            |
| $> 63$            | 4.8             |

Table 8 indicates the extent to which respondents concurred relative to eight statements. Given that the MSs are  $> 4.20 \leq 5.00$  the concurrence can be deemed to be between agree to agree strongly agree / strongly.

Table 8 indicates the importance of five parameters to respondents' organisations on a scale of 1 (not important) to 5 (very important), and a mean score (MS) ranging between 1.00 and 5.00. It is notable that all the MSs are above the midpoint score of 3.00, which indicates that in general the respondents can be deemed to perceive the parameters as important. However, given that the MSs are all  $> 4.20 \leq 5.00$ , the respondents can be deemed to perceive them to be between more than important to very important / very important.

Furthermore, it is notable that three parameters, namely blended learning decreases travel time expenses for training, respondents were satisfied with blended learning and blended learning and blended learning to be relevant on the respondents' work. Simply put, the respondents found blended learning to be something worthwhile.

**Table 4.8: Degree of concurrence relative to various statements**

| Statement                                                                             | Response (%)                         |     |     |      |      | MS  | Rank |
|---------------------------------------------------------------------------------------|--------------------------------------|-----|-----|------|------|-----|------|
|                                                                                       | Strongly disagree.....Strongly agree |     |     |      |      |     |      |
|                                                                                       | 1                                    | 2   | 3   | 4    | 5    |     |      |
|                                                                                       | SD                                   | D   | N   | A    | SA   |     |      |
| Blended learning decreases travel time expenses for training events.                  | 0.0                                  | 0.0 | 2.0 | 9.0  | 12.0 | 6.8 | 1    |
| I am satisfied with blended learning.                                                 | 0.0                                  | 0.0 | 0.0 | 5.0  | 19.0 | 4.8 | 2    |
| I find blended learning content to be relevant to my work                             | 0.0                                  | 0.0 | 1.0 | 9.0  | 13.0 | 4.5 | 3    |
| Blended learning keeps me informed about my work.                                     | 0.0                                  | 0.0 | 5.0 | 5.0  | 13.0 | 4.5 | 4    |
| A blended course helps with time management.                                          | 0.0                                  | 0.0 | 7.0 | 4.0  | 12.0 | 4.4 | 5    |
| With blended learning there is less classroom time from work (increased productivity) | 0.0                                  | 0.0 | 0.4 | 10.0 | 9.0  | 4.3 | 6    |
| Blended learning accommodates all learning types                                      | 0.0                                  | 0.0 | 5.0 | 12.0 | 6.0  | 4.2 | 7    |
| Blended learning has a positive impact on business efficiency.                        | 0.0                                  | 0.2 | 3.0 | 4.0  | 5.0  | 4.1 | 8    |

Table 4.9 indicates the importance of five parameters to respondents' blended learning on a scale of 1 (not at all important) to 5 (extremely important), and a mean score (MS) ranging between 1.00 and 5.00. It is notable that all the MSs are above the midpoint score of 3.00, which indicates that in general the respondents can be deemed to perceive the parameters as important. However, given that the MSs are all  $> 4.20 \leq$

5.00, the respondents can be deemed to perceive them to be between more than important to very important / very important. It is also notable that of all the MSs are  $> 4.20 \leq 5.00$ , which indicates that the importance of the factors is between near major to major / major. A further MSs are  $> 3.40 \leq 4.20$  - between important to more than important / more than important. Only 1 factor has a MS  $> 2.60 \leq 3.40$  - between less than important to important / important. With respect to the upper half of the MS range  $> 4.20 \leq 5.00$ , 1 factor has MSs  $> 4.60$ . Simply put, the respondents perceived blended learning as being important. The MS of 'Having access to online resources that expand the information already obtained in class' is  $> 2.60 \leq 3.40$ , which indicates the importance is between near limited to important / important.



**Table 4.9: Importance of the degree to which blended learning provides flexible training**

| Parameter                                                                                | Not.....<br>Extremely |     |     |      |     | MS  | Rank |
|------------------------------------------------------------------------------------------|-----------------------|-----|-----|------|-----|-----|------|
|                                                                                          | 1                     | 2   | 3   | 4    | 5   |     |      |
|                                                                                          |                       |     |     |      |     |     |      |
| Watch the lecture online as a video.                                                     | 1.0                   | 4.0 | 3.0 | 4.0  | 5.0 | 4.6 | 1    |
| The convenience of not having to come to class as often.                                 | 0.0                   | 0.0 | 2.0 | 3.0  | 5.0 | 4.5 | 2    |
| Accessibility to learning activities at times convenient to me                           | 0.0                   | 0.0 | 0.0 | 9.0  | 5.0 | 4.4 | 3    |
| Exploring my own area on interest.                                                       | 0.0                   | 0.0 | 0.0 | 21.0 | 2.0 | 4.1 | 4    |
| Working through the course material at my own pace.                                      | 0.0                   | 0.0 | 2.0 | 18.0 | 5.0 | 4.0 | 5    |
| The flexibility of being able to complete assignments anyplace/anytime.                  | 0.0                   | 2.0 | 6.0 | 10.1 | 5.1 | 3.9 | 6    |
| Having access to online resources that expand the information already obtained in class. | 0.0                   | 0.0 | 3.0 | 1.0  | 9.0 | 2.8 | 7    |

Table: 10 indicates the level of motivation towards blended learning. Table 9 indicates the importance of five parameters to respondents' blended learning on a scale of 1 (not at all motivating) to 5 (extremely motivating), and a mean score (MS) ranging between 1.00 and 5.00. It is notable that all the MSs are above the midpoint score of 3.00, which indicates that in general the respondents can be deemed to perceive the

parameters as motivating. However, given that the MSs are all  $> 4.20 \leq 5.00$ , the respondents can be deemed to perceive them to be between more than important to very motivating / extremely important.

**Table 4.10: Level of motivation towards blended learning**

| Factor                                        | Response (%) |                           |     |     |     | M<br>S | Rank |   |
|-----------------------------------------------|--------------|---------------------------|-----|-----|-----|--------|------|---|
|                                               |              | Not<br>.....Extrem<br>ely |     |     |     |        |      |   |
|                                               |              | 1                         | 2   | 3   | 4   |        |      | 5 |
| Interaction with other learners and trainers. |              | 0.0                       | 0.0 | 0.0 | 8.0 | 15.0   | 4.7  | 1 |
| Availability of instructional videos.         |              | 0.0                       | 2.0 | 4.0 | 1.0 | 16.1   | 4.6  | 2 |
| Sense of community.                           |              | 1.0                       | 1.0 | 1.0 | 8.0 | 12.0   | 4.5  | 3 |

Table 11 indicates the degree to which barriers affect respondents towards using blended on a scale of 1 (not affected) to 5 (extremely affected), and a mean score (MS) ranging between 1.00 and 5.00. It is notable that all the MSs are above the midpoint score of 3.00, which indicates that in general the respondents can be deemed to perceive the parameters as motivating. However, given that the MSs are all the MS of barriers is  $> 2.60 \leq 3.40$ , which means that the concurrence is between disagree and neutral. The barriers are the impediment to blended learning and interventions are needed.

**Table 4.11: The degree to which barriers affect respondents towards using blended learning**

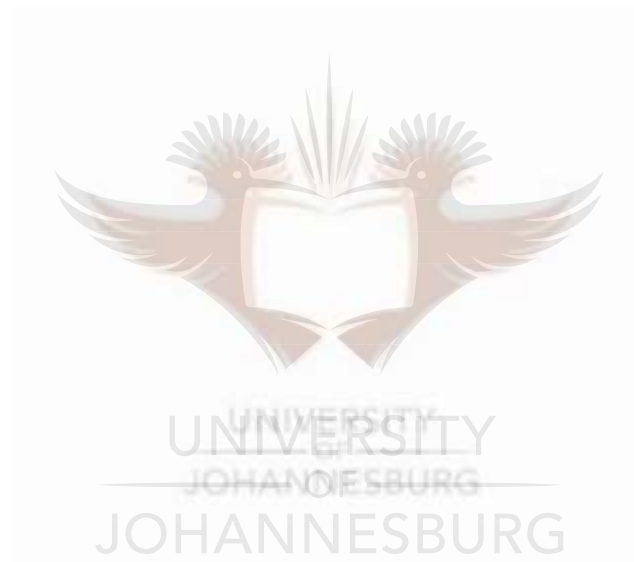
| Parameter                            | Not.....<br>Extremely |      |      |     |     | MS   | Rank |
|--------------------------------------|-----------------------|------|------|-----|-----|------|------|
|                                      | 1                     | 2    | 3    | 4   | 5   |      |      |
|                                      |                       |      |      |     |     |      |      |
| Experience of using the course tools | 7.0                   | 3.0  | 12.0 | 4.0 | 5.0 | 2.92 | 1    |
| Lack of internet skills              | 9.0                   | 7.0  | 6.0  | 1.0 | 0.0 | 2.37 | 2    |
| Availability of instructional videos | 6.0                   | 11.0 | 5.0  | 1.0 | 1.0 | 2.36 | 3    |
| Anxiety                              | 11.0                  | 7.0  | 3.0  | 2.0 | 0.0 | 2.33 | 4    |
| Connectivity                         | 13.0                  | 8.0  | 0.0  | 1.0 | 1.1 | 2.14 | 5    |
| Fear of technology                   | 10.0                  | 8.0  | 4.0  | 1.0 | 0.0 | 1.95 | 6    |

#### **4.1.3 OPEN ENDED QUESTIONS**

Respondents were asked to give their views on blended learning. The following are a sample of their answers:

- The organisation should provide employees with data.
- The learner management system should be simple.
- The organisation should improve connectivity.
- The system should be easy to explore.
- The organisation should adopt the culture of learning with technology.
- Blended learning should be rolled out to all training programs.
- Learner management should be fit for purpose.
- Invest in learner management system.

- The computer system must be fixed we must also be allowed to practice the system to get used to it.



## **CHAPTER 5. SUMMARY, DISCUSSION, RECOMMENDATIONS, CONCLUSIONS AND FINAL WORD**

The workplace environment is rapidly changing because of emerging technologies and access to the internet. Changes in technology, coupled with what is now termed the new normal, interrupted how society had positioned itself in both the way it lives and educates itself. The aim of this study was to determine how the use of blended teaching can improve workplace training in the transport sector. The study focussed in the railway training and development environment. Questionnaires used provided a valuable source of data and enabled a clearer understanding into workplace use of blended teaching platforms. Participant reacted positively in the research and have an interest in the outcome of this study.

This study summarized the study in chapter 1. Chapter 2 discussed the literature pertaining to the use of blended teaching in the workplace to improve workplace training in the transport sector. The literature concentrated on the following areas:

- The revolution of blended learning;
- The origin of blended learning;
- Using the world wide web for learning;
- The need for blended learning;
- The need for blended learning: South African perspective;
- Greater access to knowledge and the need for blended learning;
- Advances in technology and the need for blended learning;
- Benefits of blended learning;
- Limitations of blended learning; and
- Possible Solutions for blended learning implementation.

Chapter 3 incorporated the details of the research design and methodology. This chapter provided justification of the method used and systematic details regarding sampling, data collection, data recording, and data analysis. This chapter also briefly described the participants and ethical considerations. Chapter 4 presented the findings that emerged as themes from the data collected. The findings presented in this study

resulted from coding through data analysis. Chapter 5 discusses the summary and recommendations and concludes the research.

## **5.1 LIMITATIONS OF THE STUDY**

The major limitation of this research is that the researcher could only survey PRASA and Transnet. Major challenge encountered by the researcher is in data collection. The researcher had to send had to remind the participants through their managers. Although the online version of the questionnaires was earlier than the emailed ones, participants also had to be reminded. Findings of the study are therefore generalised and exclusive to these two big railway operators in the transport sector. It is recommended to expand the sample size in order to include other transport operators in future studies.

The participants, important members of the faculty, are always busy; thus, there was a delay in retrieving the questionnaires. The same experience occurred with the online version of the questionnaire; researchers had to send a reminder and waited for some time to collect the required data. The primary method of data collection was survey questionnaires, which can be administered and evaluated quickly with no need to spend time at the organisation prior to administering the survey, and the responses can be tabulated within a short timeframe (Choy, 2014). The questionnaire was deliberate in directing responses towards the use of blended teaching to improve workplace training in the transport sector. Future research can adopt a mixed method research to include qualitative surveys. This will extend the research area to include a larger sample size as well as provide the researcher with statistical information on the topic.

## **5.2 CONCLUSIONS EMANATING FROM THE FINDINGS**

Chapter 4 presented the findings from the data collected through the survey questionnaire. The findings arose, stating the perception towards flexibility, motivations and barriers to participant's use of blended learning in workplace training. This section will present the conclusions emanating from the findings.

### **5.2.1     *Impact of blended learning on business efficiency***

Through the questionnaire, all participants except one train driver and one metro guard agree that blended learning has a positive impact on business efficiency. All 23 participants the respondents found blended learning to be something worthwhile.

### **5.2.2     *Flexibility and timely training***

At present, some train drivers and metro guards are working at remote depots (out of the city depots). Remote depots have bad or no connectivity infrastructure. They rely on phone data or Wi-Fi routers to connect and/or download videos. Two train drivers from remote depots and one train control officer do not see the importance of watching the lecture online as a video. All participants see the convenience of not have to go to class as often to be extremely important.

Access to ICT resources is important to all participants. Accessibility to learning activities at time of convenience has been observed to be important. Poor connectivity and lack of resources affect the mixing of traditional face-to-face training and online learning. Lack of connectivity can slow down the pace of training. It is disappointing that remote depots seemed to be neglected and staff was left to fend for itself. ICT department should improve connectivity in the remote depots or issue everyone with WI-FI router. It emerged that one of the important part of blended learning is having access to online resources that expand the information already obtained in class.

### **5.2.3     *Level of motivation towards blended learning***

Poon (2013) stated that enhancing students' motivation to learn on their own "at their own pace and in their own time" is an important aspect of a blended learning environment. According to Ibrahim and Nat (2019) motivation has to do with energy and all other aspects of initiation and intention and further posit that Motivation is also regarded as a product of the interaction between a person and a particular situation and is not a personal trait. Interaction motivates the participants with other learners and trainers, having a sense of community and the ability of instructional videos.

#### **5.2.4 *Barriers towards the use of blended learning***

The study found four themes that inductively make up barriers to using blended learning. These are lack of internet skills, fear of technology, availability of instructional videos, and experience in using the course tools, connectivity and anxiety, respectively. Antwi-Boampong (2021) found that some barriers to implementation of blended learning include a lack of institutional support from top management commitment. The finding reveals challenging areas such as power interruptions (load shedding), internet connectivity failure and lack of internet skills as the fundamental barriers in blended learning. The barriers are the impediment to blended learning and interventions are needed.

### **5.3 RECOMMENDATIONS FOR FUTURE STUDIES**

This study presents the perceptions of railway transport sector regarding their use of blended teaching to improve workplace teaching in the transport sector. Blended learning has been adopted to a certain degree in the transport sector. Its effectiveness can be realised when it is accepted and adopted by workplace training managers, trainers and learners. From the findings and the limitations of this study, future research on the use of blended teaching to improve workplace training in the transport sector is suggested. The following are recommended for future research:

- Extend the research population to include additional transport operators such as truck, buses and taxis.
- Increase the sample size
- Combine research methods as far as is workable (Include quantitative research to develop a mixed method research)

## 5.4 CONCLUSION

It can be concluded that a blended learning approach can strengthen the training process, SA railway like any other industry, find itself in a developing knowledge-based world where the skills require constant update.

It is recommended that BL should be implemented in the following manner:

- Consult widely with stakeholders
- Create and/or review BL guidelines.
- Assess needs and capabilities
- Ensure practical application and relevance
- Create a community of educational developers.
- Make lessons and materials accessible on-demand
- Develop competencies with existing technologies (e.g. devices such as phones)
- Provide adequate contact time, either face-to-face or online, with students.
- Don't force an unwanted program on employees
- Familiarise end-users with the tools they'll need to access and take part in training.
- Fit in training into the flow of work.

## 5.5 A FINAL WORD

Training in the workplace results in employees gaining the skills to perform their duties optimally. A person who is skilled at his or her duty can do the work. This person relies not only on the skill, knowledge is also necessary. A person gains this knowledge know and understands what he or she is doing. A combination of knowledge and skill means a person will know what to do. In the railway environment, employees rely on unique method to operate trains. There is manual authorisation and technological authorisation of trains. When technology fails a manual authorisation is activated. A combination of technology and manual operation is required. This paragraph illustrates the important of blending the new with the old.

Blended teaching can achieve quite a lot in the transport sector. Technology is coming to the railway in full force. New train and the signal systems are bringing technology into the system. Learners in this environment can learn at their own pace using train or signal simulators. Train simulator the use of computer-generated images using visual reality is the way to go. This blend of learning will mean learners are no longer confined to class.

The railway environment is a highly unionised environment and implementing blended teaching will need proper consultation with all stakeholders to avoid rejection. Learning specialist in the workplace should be futuristic in their planning for training. They should design a curriculum with blended learning in mind. This study will hopefully spark interest in others to conduct research in the transport sector, especially the South African railway perspective.

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## Appendix A: Research request letter

Re: REQUEST TO CONDUCT RESEARCH INTERVIEWS IN YOUR ORGANISATION

I am a Masters student at the University of Johannesburg currently writing my Dissertation entitled: **The use of blended teaching to improve workplace training in the transport sector**. The study is undertaken through the supervision of Prof. G Lautenbach, Faculty of Education, University of Johannesburg.

I have selected your organisation as a possible option for my research study. Participation in this survey is voluntary and your responses will be kept confidential and anonymous. You may choose not to participate without fear. I assure you that neither the names of the participants will be published in the research report. Upon acceptance of this request, the comprehensive Ethical Clearance forms will be forwarded. For the research, I intend to conduct a survey with Train drivers, managers, metro guards, senior technical training officers, train control officers and yard officials and it should not take more than 40 minutes.

I request your permission to conduct the research in your organisation. I am looking forward to have a favourable response. Your consideration in this matter will be highly appreciated. If you have any questions concerning this study, please contact me (contact details below).

Yours sincerely



Mboneni Stanely Mulaudzi

083 230 6000

## Appendix B: Research Consent letters

[www.prasa.com](http://www.prasa.com)



**prasa**  
PASSENGER RAIL AGENCY  
OF SOUTH AFRICA

PRASA HOUSE  
1000 Burnwell Street  
Ficksburg  
Ficksburg

Private Bag 2011  
Braamfontein, 2019  
t: +27 11 745 9032

Ref.: Research permissions/2020

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TO WHOM IT MAY CONCERN

**Permission to conduct research at the Passenger Rail Agency of South Africa (PRASA) environment for academic purposes - Student: Mr M.S. Mulaudzi**

Research / Thesis topic: *"Exploring the use of blended learning to improve business efficiency at a railway institution."*

Permission is hereby granted to Mr. M.S. Mulaudzi to conduct research at PRASA for academic purposes as part of his studies to complete an MEd programme in ICTs (Education) at the University of Johannesburg, subject to honouring the following agreement:

**"Confidentiality Agreement – for the purpose of conducting research in the PRASA environment and/or using Company information for research purposes"**

Mr. Mulaudzi has signed the above Confidentiality Agreement with PRASA and agreed that the above research is for academic purposes only and that no part of the research will be published without written approval from PRASA.

Yours faithfully



Portia Tsolekile  
Manager: Group Strategy  
Passenger Rail Agency of South Africa  
E-mail: [ptsolekile@prasa.com](mailto:ptsolekile@prasa.com)  
Date: 31 January 2020

## Appendix C: Confidentiality agreement

www.prasa.com

Be moved



**prasa**  
PUBLIC REVENUE AND  
INFRASTRUCTURE

### CONFIDENTIALITY AGREEMENT

- for the purpose of conducting research within PRASA environment, or using Company information for research purposes

I, the undersigned

1. herewith undertake that all information disclosed or submitted, either orally, in writing or in other tangible or intangible form by PRASA, (its subsidiaries, business units, its employees, agents and/or consultants) to me, or made available to me, or details of PRASA's business or interest of which I may become aware of in respect of the research being done by myself for study purposes at *University of Johannesburg* (University/College), to keep confidential and not to divulge to anyone either privately or publicly for which PRASA did not give written consent;
2. guarantee that I will apply the information, detail or knowledge in clause 1 only for the purpose of my academic research;
3. indemnify PRASA against any claims that may be instituted against it, amounts that may be claimed or losses that PRASA may suffer in consequence of a violation by me of any provision included in this agreement;
4. agree that the provisions of this agreement binds me to PRASA, even if I cease to be a student, employee, representative or advisor of the *University of Johannesburg* (University/College), depending as the case may be after ceasing to be such a person.
5. shall immediately disclose in writing all new information in my possession or under my care relating to the research, provided that such new information must have been developed during the course of the research relating to this agreement.
6. agree that PRASA will have a final say on whether my final work gets published, either in journals, university libraries or any arena where such work may be accessed either electronically or physically. I further accept that PRASA reserves the rights to put limitations on which parts of my work may be published, either in full or in sections. This clause is not applicable to the information that is in the public domain, and/or published in the Annual Reports.

RCUS\_00100-PRASA Page 1 of 2

## Appendix D: REC application letter



To whom it may concern

**MS Mulaudzi – MEd research at UJ**

This is to confirm that Mboneni Stanely Mulaudzi, an employee of PRASA Rail (SAP Number 2002 3137) is currently registered for the MEd in ICTs in Education. He aims to collect primary data for his research at PRASA Rail. The study is titled "Exploring the use of blended learning to improve business efficiency at a railway institution"

This is a low risk study and we foresee no ethical issues. Please feel free to contact me at any time if you need to find out more about the study

Yours sincerely



Geoff Lautenbach  
28/01/2020

=====xxXX=====

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## Appendix E: Research questionnaire

This questionnaire has been designed to solicit information for purely academic purposes. It is intended to assist the researcher, in partial fulfillment of the award of a Med Programme in ICTs (Education). It is worth mentioning that the data collected shall be used exclusively for academic purposes.

Please tick the answer that best represents you

1. What is your job title?

Train Driver

2. How old are you?

|         |  |         |  |         |  |         |  |
|---------|--|---------|--|---------|--|---------|--|
| 21 - 30 |  | 31 - 40 |  | 41 - 50 |  | 51 - 63 |  |
|---------|--|---------|--|---------|--|---------|--|

3. How long have you been working for PRASA

|        |  |         |  |         |  |         |  |
|--------|--|---------|--|---------|--|---------|--|
| 1 - 10 |  | 11 - 20 |  | 21 - 30 |  | 31 - 44 |  |
|--------|--|---------|--|---------|--|---------|--|

### Section 1

Please indicate your level of agreement with the following statements:

| Statement                                                            | Strongly disagree | Disagree | Neither agree nor disagree | Agree | Strongly Agree |
|----------------------------------------------------------------------|-------------------|----------|----------------------------|-------|----------------|
| Blended learning decreases travel time expenses for training events. |                   |          |                            |       |                |
| I find blended learning content to be relevant to my work.           |                   |          |                            |       |                |
| Blended learning has a positive impact on business efficiency.       |                   |          |                            |       |                |
| Blended learning accommodates all learning types.                    |                   |          |                            |       |                |

|                                                                                              |  |  |  |  |  |
|----------------------------------------------------------------------------------------------|--|--|--|--|--|
| Blended learning keeps me informed about my work.                                            |  |  |  |  |  |
| A blended course format helps with time management.                                          |  |  |  |  |  |
| With blended learning, there is less classroom time away from work (Increased productivity). |  |  |  |  |  |
| I am satisfied with blended learning.                                                        |  |  |  |  |  |

## Section 2

Please indicate the importance of the degree to which blended learning provides flexible and timely training to enhance an employee learning program in the transport sector:

| Statement                                                                                | Not at all important | Slightly important | Moderate important | Very important | Extremely important |
|------------------------------------------------------------------------------------------|----------------------|--------------------|--------------------|----------------|---------------------|
| Accessibility to learning activities at times convenient to me.                          |                      |                    |                    |                |                     |
| Working through the course material at my own pace.                                      |                      |                    |                    |                |                     |
| The convenience of not having to come to class as often.                                 |                      |                    |                    |                |                     |
| Exploring my own area of interest.                                                       |                      |                    |                    |                |                     |
| Having access to online resources that expand the information already obtained in class. |                      |                    |                    |                |                     |
| Watch the lecture online as a video.                                                     |                      |                    |                    |                |                     |
| The flexibility of being able to complete assignments anyplace/anytime                   |                      |                    |                    |                |                     |

### Section 3

Please indicate your level of motivation towards blended learning:

| Statement                                     | Not at all motivating | Slightly motivating | Undecided | Very motivating | Extremely motivating |
|-----------------------------------------------|-----------------------|---------------------|-----------|-----------------|----------------------|
| Interaction with other learners and trainers. |                       |                     |           |                 |                      |
| Sense of community                            |                       |                     |           |                 |                      |
| Availability of instructional videos.         |                       |                     |           |                 |                      |

### Section 4

Please indicate the degree to which these barriers affect you towards using blended learning:

| Statement                             | Not affected | Slightly affected | Neutral | Very affected | Extremely affected |
|---------------------------------------|--------------|-------------------|---------|---------------|--------------------|
| Lack of internet skills.              |              |                   |         |               |                    |
| Fear of technology                    |              |                   |         |               |                    |
| Availability of instructional videos. |              |                   |         |               |                    |
| Experience of using the course tools. |              |                   |         |               |                    |
| Connectivity                          |              |                   |         |               |                    |
| Anxiety                               |              |                   |         |               |                    |

Please suggest ways and means to adopt blended learning to improve the business efficiency of a transport sector?

Thank you.

